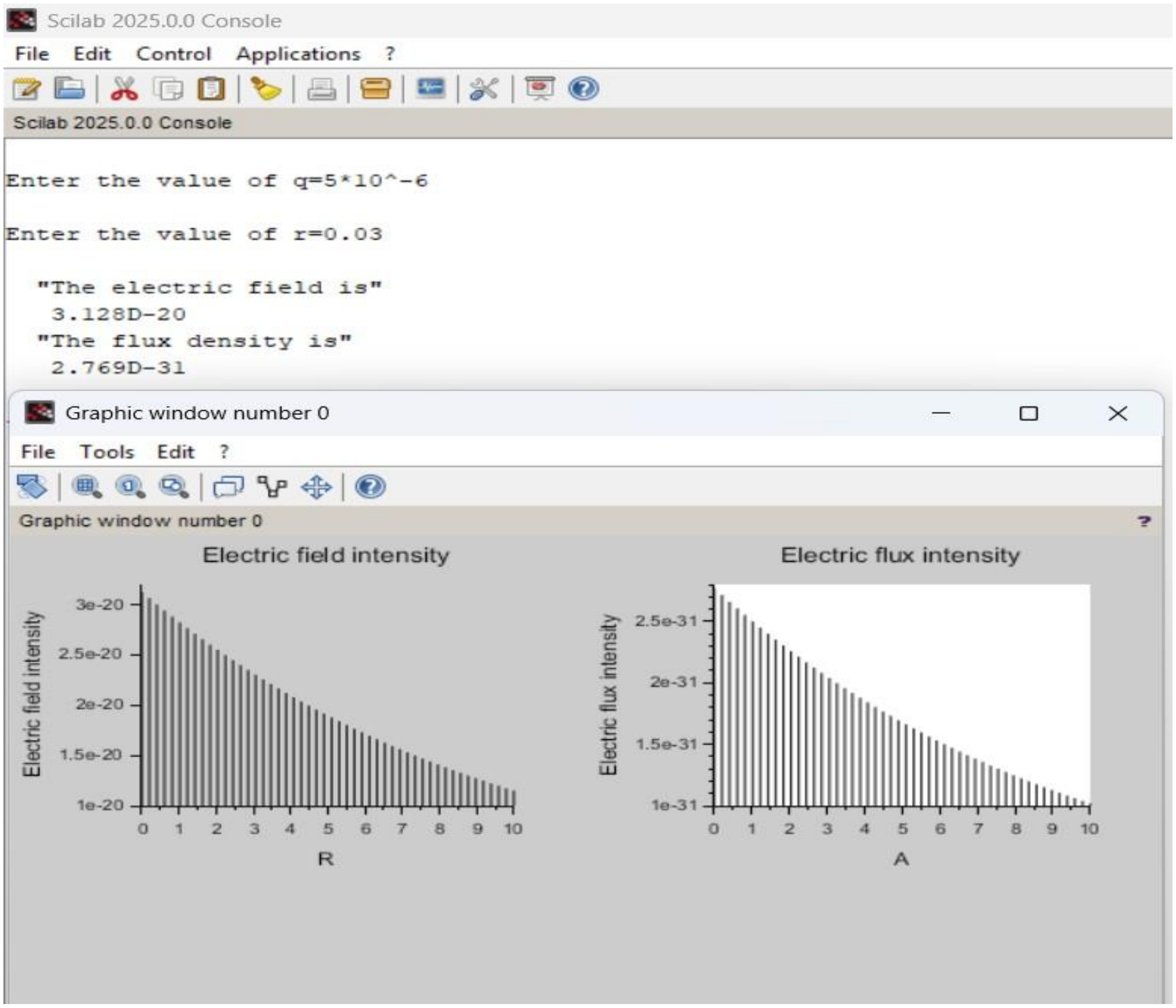
Exp2

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

PROGRAM:

```
clc; clear;
q=input('Enter the value of q=');
r=input('Enter the value of r=');
efcilon=8.854*10^-12;
pi=3.14
E=q/4*pi*efcilon*r^2;
disp("The electric field is", [E]);
D=E*efcilon;
disp("The flux density is", [D]);
x = linspace(0, 10, 50);
y = E * exp(-0.1 * x);
z = D * exp(-0.1 * x); figure();
subplot(2, 2, 1);
plot2d3(x,y);
xlabel("R");
ylabel("Electric field intensity");
title('Electric field intensity');
subplot(2, 2, 2);
plot2d3(x,z);
xlabel("A");
ylabel("Electric flux intensity");
title('Electric flux intensity');
```

Output:



Calculation:

Exp:- 2

$Q = 5\times10^{-6}$

$r = 3\text{ cm} \Rightarrow 3\times10^{-2}\text{ m} = 0.03\text{ m}$

$\varepsilon = 8.854\times10^{-12}$

⇒ Electric field ~~flux Density~~ (E)

$$\boxed{E = \frac{Q}{4\pi r^2 \varepsilon}}$$

$$= \frac{5\times10^{-6}}{4\pi\times(0.03)^2} = \frac{5\times10^{-6}}{4\pi\times9\times10^{-4}\times\varepsilon} = \frac{5\times10^{-6}}{1.1309\times10^{-2}\,\varepsilon}$$

$$E = \frac{5\times10^{-6}}{1.001\times10^{-13}} = 3.128\times10^{-20}$$

⇒ Electric flux density

$$\boxed{D = E\cdot\varepsilon}$$

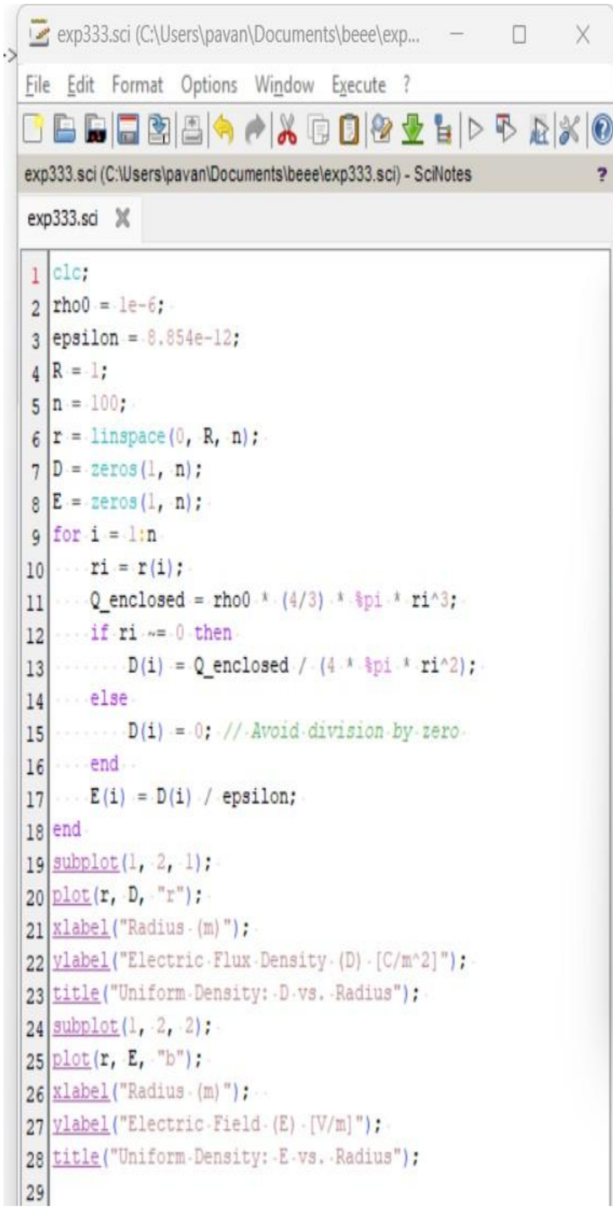
$$= 3.128\times10^{-20}\times8.854\times10^{-12}$$

$$D = 2.769\times10^{-31}$$

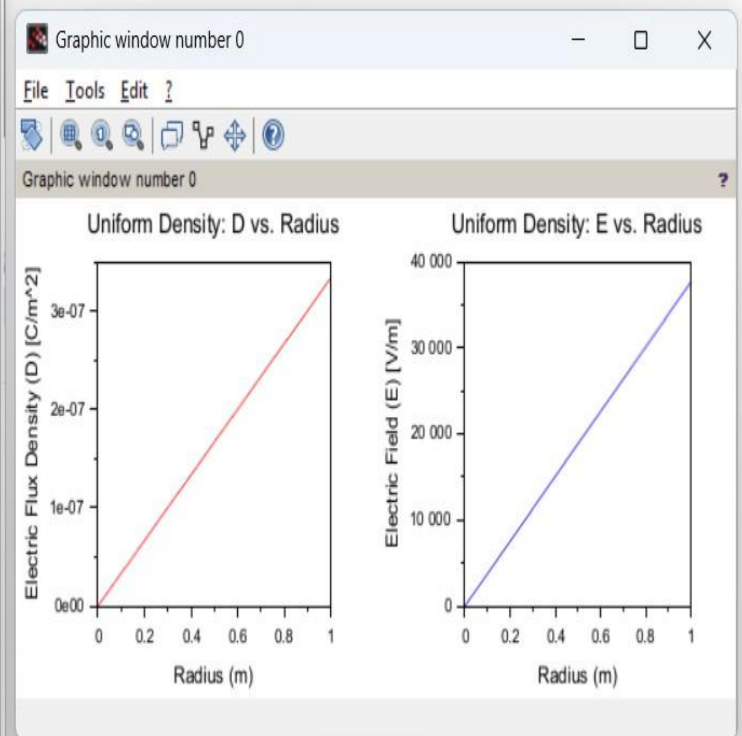
O/P

TEST CASE: 1

OUTPUT:

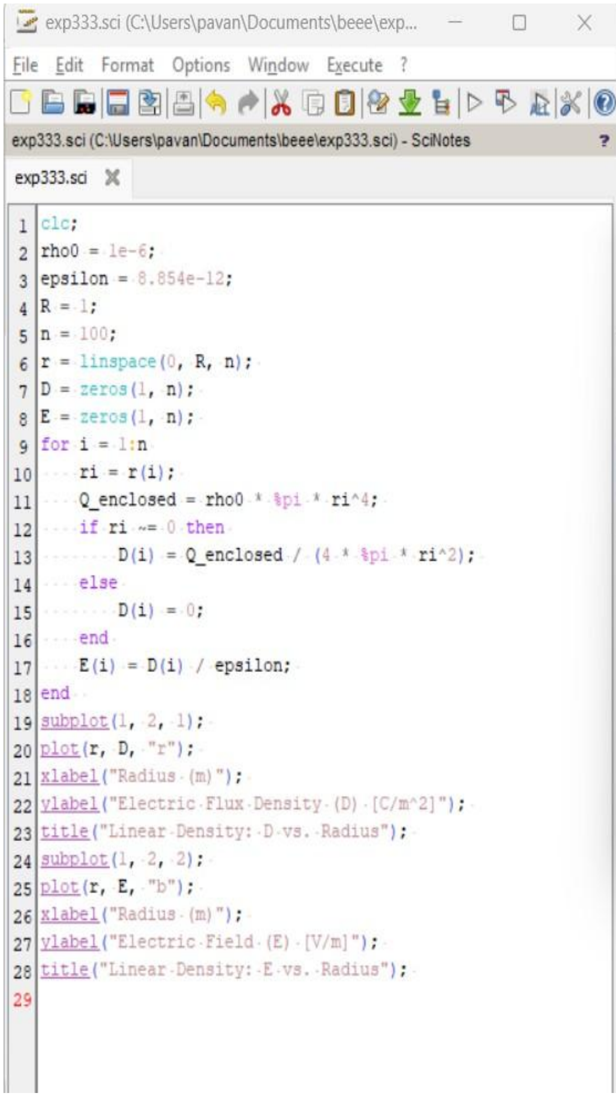



```
clc;
rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;
n = 100;
r = linspace(0, R, n);
D = zeros(1, n);
E = zeros(1, n);
for i = 1:n
    ri = r(i);
    Q_enclosed = rho0 * (4/3) * %pi * ri^3;
    if ri ~= 0 then
        D(i) = Q_enclosed / (4 * %pi * ri^2);
    else
        D(i) = 0; // Avoid division by zero
    end
    E(i) = D(i) / epsilon;
end
subplot(1, 2, 1);
plot(r, D, "r");
xlabel("Radius (m)");
ylabel("Electric Flux Density (D) [C/m^2]");
title("Uniform Density: D vs. Radius");
subplot(1, 2, 2);
plot(r, E, "b");
xlabel("Radius (m)");
ylabel("Electric Field (E) [V/m]");
title("Uniform Density: E vs. Radius");
```

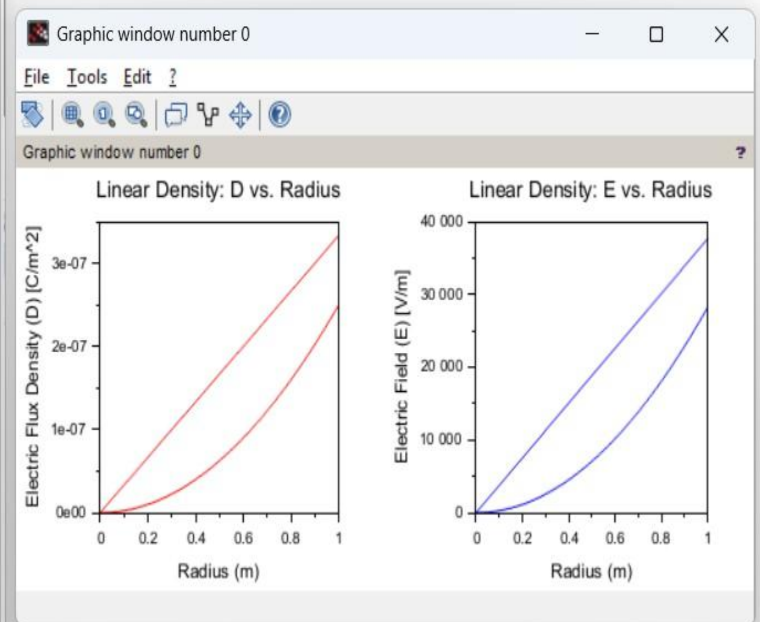


TEST CASE: 2

OUTPUT:


exp333.sci (C:\Users\pavan\Documents\beee\exp...
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci

```
clc;
rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;
n = 100;
r = linspace(0, R, n);
D = zeros(1, n);
E = zeros(1, n);
for i = 1:n
    ri = r(i);
    Q_enclosed = rho0 * %pi * ri^4;
    if ri ~= 0 then
        D(i) = Q_enclosed / (4 * %pi * ri^2);
    else
        D(i) = 0;
    end
    E(i) = D(i) / epsilon;
end
subplot(1, 2, 1);
plot(r, D, "r");
xlabel("Radius (m)");
ylabel("Electric Flux Density (D) [C/m^2]");
title("Linear Density: D vs. Radius");
subplot(1, 2, 2);
plot(r, E, "b");
xlabel("Radius (m)");
ylabel("Electric Field (E) [V/m]");
title("Linear Density: E vs. Radius");
```



TEST CASE: 3

OUTPUT:

exp333.sci (C:\Users\pavan\Documents\beee\exp...

File Edit Format Options Window Execute ?

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

exp333.sci

```
clc;
rho0 = 3e-6;
epsilon = 8.854e-12;
R = 2;
n = 200;
r = linspace(0, 4, n);
D = zeros(1, n);
E = zeros(1, n);
Q_total = rho0 * (4 * %pi * R^3 / 3);
for i = 1:n
    ri = r(i);
    if ri < R then
        Q_enclosed = rho0 * (4 * %pi * ri^3 / 3);
    else
        Q_enclosed = Q_total;
    end
    D(i) = Q_enclosed / (4 * %pi * ri^2);
    E(i) = D(i) / epsilon;
end
subplot(1, 2, 1);
plot(r, D, "r");
xlabel("Radius (m)");
ylabel("Electric Flux Density (D) [C/m^2]");
title("Uniform Charge Density: D vs. Radius");
subplot(1, 2, 2);
plot(r, E, "b");
xlabel("Radius (m)");
ylabel("Electric Field (E) [V/m]");
title("Uniform Charge Density: E vs. Radius"); clc;
rho0 = 3e-6;
epsilon = 8.854e-12;
R = 2;
n = 200;
r = linspace(0, 4, n);
D = zeros(1, n);
```